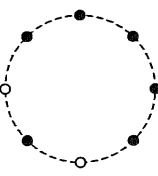


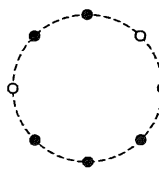
Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

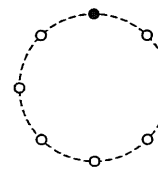
1. [ACJC/H2/Prelims 2018/P1/19]
A total of eight electrons (O) and protons (●) are spaced evenly along the circumference of a circle as shown. Which of the combinations has the smallest electric field strength at the centre of the circle?
- A



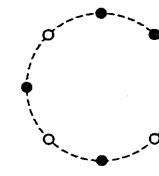
B

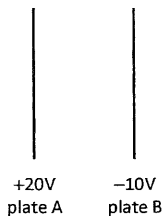


C



D


2. [ACJC/H2/Prelims 2018/P1/20]
Two long parallel plates, spaced a distance of 30 cm apart, have electric potentials of +20 V and −10 V as shown.



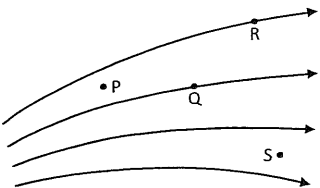
- Which statement is correct?
- A At any point which is 20 cm to the right of plate A, the work done by an external agent per unit charge to bring a small positive test point charge from infinity to that point is non-zero.
- B The electric field strength between the two plates is 1 V m^{-1} , and it is directed towards plate B.
- C An electron placed at 10 cm to the left of plate B will gain 20 eV of electric potential energy when it reaches plate A.
- D When the plates are connected to the ends of a resistor, the electric field strength within the two plates will become zero.
3. [AJC/H2/Prelims 2018/P1/21]
A particle of charge $+3e$ remains at rest midway between a pair of horizontal, parallel plates having a separation of 15 mm. The potential difference between the plates is 660 V.
What is the weight of the particle?
- A $2.1 \times 10^{-14} \text{ N}$

B $1.1 \times 10^{-14} \text{ N}$

C $2.1 \times 10^{-17} \text{ N}$

D $1.1 \times 10^{-17} \text{ N}$

4. [CJC/H2/Prelims 2018/P1/17]
A region of electric field is represented by the electric field lines shown. P, Q, R and S are 4 points in the field.

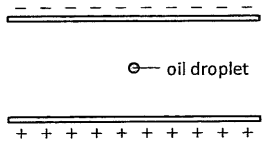


- Which of the following statements is incorrect?
- A The electric force on a charged particle is stronger when it is located at P than that at R.
- B A positively charged particle released from rest at Q will travel along the electric field line which passes through Q.
- C The electric potential energy of a negatively charged particle at S is higher than that at P.
- D The electric field strength at Q is stronger than that at S.
5. [CJC/H2/Prelims 2018/P1/18]

An isolated metal sphere of radius 0.1 m is positively charged. A small charge is brought from a distant point to a distant 0.5 m from the centre of the metal sphere. The work done against the electric field is W and the electric force on the charge at its final position is F .
What is the work done against the electric field and the electric force on the charge, if the charge has been brought to a distant 1.0 m from the centre of the metal sphere?

	work done against electric field	force on charge at its final position
A	$\frac{W}{4}$	$\frac{F}{4}$
B	$\frac{W}{4}$	$\frac{F}{2}$
C	$\frac{W}{2}$	$\frac{F}{4}$
D	$\frac{W}{2}$	$\frac{F}{2}$

6. [DHS/H2/Prelims 2018/P1/19]
A positively charged oil droplet is held stationary in an electric field of strength E as shown below.



- A different oil droplet of the same density is held stationary in an electric field of different strength. What is this electric field strength if the oil droplet has half the charge and twice the radius of the original droplet?
- A $2E$

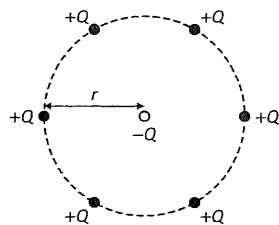
B $4E$

C $8E$

D $16E$

7. [DHS/H2/Prelims 2018/P1/20]

A negative point charge is surrounded symmetrically by six positive point charges at distance r as shown.

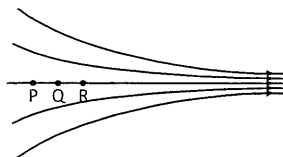


How much work is done by the forces of attraction when the point charge at the centre is removed to infinity?

- A $-\frac{6Q^2}{4\pi\epsilon_0 r}$ B $+\frac{6Q^2}{4\pi\epsilon_0 r}$ C $-\frac{6Q^2}{4\pi\epsilon_0 r^2}$ D $+\frac{6Q^2}{4\pi\epsilon_0 r^2}$

8. [EJC/H2/Prelims 2018/P1/17]

The diagram below shows electric field lines with points P, Q and R lying on one of the field lines. The distance PQ is equal to the distance QR.

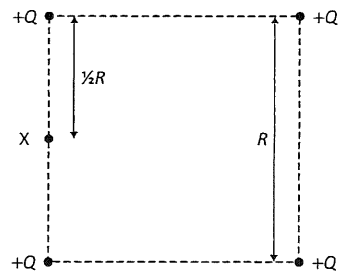


If the potentials at P and Q are 0 V and -200 V, respectively, which of the following is a possible value of the potential at R?

- A -450 V B -400 V C $+200$ V D $+250$ V

9. [EJC/H2/Prelims 2018/P1/18]

Four identical point charges are arranged at the corners of a square of length R as shown.



What is the magnitude of the electric field strength E and the electric potential V at point X?

	E	V
A	$\frac{12}{5} \cdot \frac{Q}{\pi\epsilon_0 R^2}$	$\frac{2}{5} \cdot \frac{Q}{\pi\epsilon_0 R}$
B	$\frac{12}{5} \cdot \frac{Q}{\pi\epsilon_0 R^2}$	$\left(1 + \frac{1}{\sqrt{5}}\right) \cdot \frac{Q}{\pi\epsilon_0 R}$
C	$\frac{4}{\sqrt{5}^3} \cdot \frac{Q}{\pi\epsilon_0 R^2}$	$\frac{2}{5} \cdot \frac{Q}{\pi\epsilon_0 R}$
D	$\frac{4}{\sqrt{5}^3} \cdot \frac{Q}{\pi\epsilon_0 R^2}$	$\left(1 + \frac{1}{\sqrt{5}}\right) \cdot \frac{Q}{\pi\epsilon_0 R}$

10. [HCI/H2/Prelims 2018/P1/18]

The figure below shows two charged oil drops X and Y of masses $2m$ and m , respectively, which are just prevented from falling under gravity by the application of a constant potential difference between the two parallel metal plates.

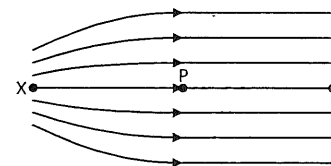


Which of the following correctly describes what will happen if the plates are moved further apart?

- A Both X and Y will move up with the same acceleration.
 B Both X and Y will move down with the same acceleration.
 C X will begin to move down with an acceleration greater than that of Y.
 D X will begin to move down with an acceleration smaller than that of Y.

11. [HCI/H2/Prelims 2018/P1/19]

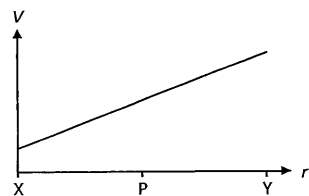
The diagram below shows the electric field lines in a region of space.



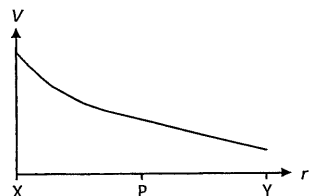
Which of the following diagrams shows the variation with distance r of the potential V along the line XY?

A

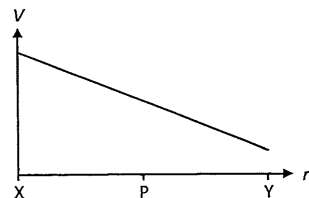
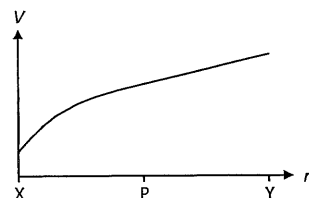
B



C

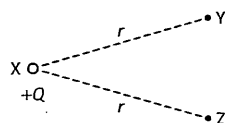


D



12. [JJC/H2/Prelims 2018/P1/19]

The figure below shows a particle of charge $+Q$ at X. The points Y and Z are equidistant from X.

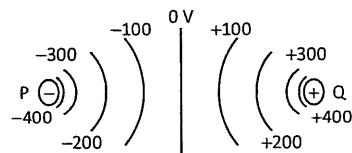


Which one of the following statements with reference to the above figure is incorrect?

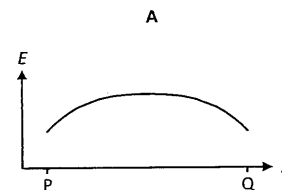
- A No work is done in taking a charge from Y to Z.
- B The electric field strength at Z acts along ZX in the direction of Z to X.
- C The magnitude of the electric field strength at Y is equal to the magnitude of the electric field strength at Z.
- D The potentials at Y and Z are equal.

13. [JJC/H2/Prelims 2018/P1/14 Modified]

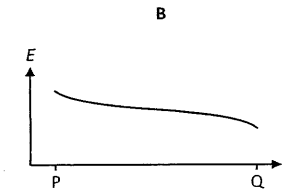
An object with a negative charge is placed at P and a similar object with a positive charge is placed at Q. The diagram shows the equipotential lines between these two objects.



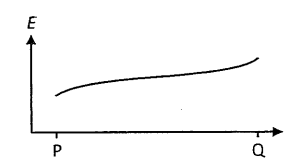
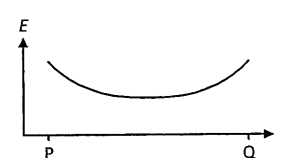
Which graph shows the variation with distance x along line PQ of the magnitude of the electric field strength E ?



C

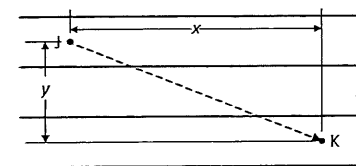


D



14. [JJC/H2/Prelims 2018/P1/15]

A small negative charge ($-Q$) is placed at point J inside a uniform electric field of field strength E . It is then moved from point J to point K.

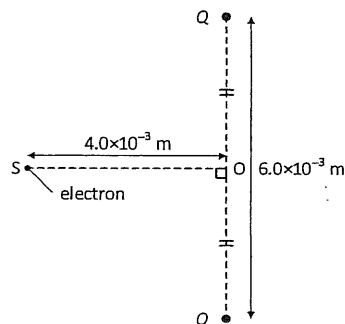


What is the change in electric potential energy of the charge?

- A gain of QEx
- B loss of QEx
- C gain of $QE\sqrt{x^2 + y^2}$
- D loss of $\frac{Q}{4\pi\epsilon_0\sqrt{x^2 + y^2}}$

15. [MI /H2/Prelims 2018/P1/22 Modified]

Two identical point charges $Q = +1.45 \times 10^{-15}$ C each are fixed 6.0×10^{-3} m apart. An electron is then projected with a speed v into the plane of the paper at point S which is 4.0×10^{-3} m away from the perpendicular bisector between the two charges.



Which of the following gives the magnitude of resultant force on the electron and the speed v with which the electron must be projected into the plane of the paper at S to just perform circular motion with its centre at point O.

	resultant force/ N	speed v / m s^{-1}
A	8.3×10^{-20}	1.2×10^4
B	1.3×10^{-19}	2.4×10^4
C	8.3×10^{-20}	2.4×10^4
D	1.3×10^{-19}	1.2×10^4

16. [MJC/H2/Prelims 2018/P1/17]

Which of the following statements about the electric potential at a point is correct?

- A The potential is given by the rate of change of electric field strength with distance.
- B The potential is equal to the work done per unit positive charge by an external force in bringing a small test charge from infinity to that point.
- C Two points in an electric field are at the same potential when a small positive charge placed along the line joining them remains stationary.
- D An alternative unit for electric potential is J m^{-1} .

17. [MJC/H2/Prelims 2018/P1/18]

The electric potentials V measured at distances x from point P to Q along a straight line are shown below:

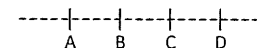
x / m	0.020	0.030	0.040	0.050	0.060
V / V	13	15	18	21	23

The electric field at $x = 0.040$ m is approximately

- A 300 V m^{-1} towards Q
- B 300 V m^{-1} towards P
- C 450 V m^{-1} towards Q
- D 450 V m^{-1} towards P

18. [NJC/H2/Prelims 2018/P1/18]

A, B, C and D are four points on a straight line as shown. A point charge $+Q$ is fixed at A and a point charge $-Q$ is moved from B to C.

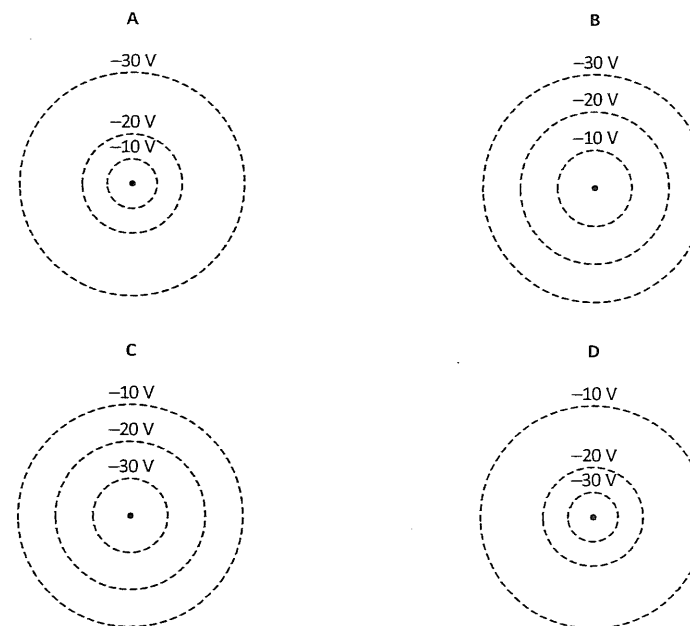


Which of the following statements is false?

- A The electric potential at point D increases.
- B The magnitude of the electric field strength at point D increases.
- C The electric potential energy of the system of two charges increases.
- D The electric field strength at the midpoint between point A and point B decreases.

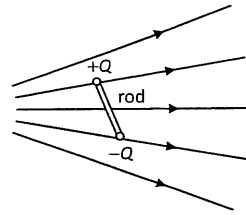
19. [NJC/H2/Prelims 2018/P1/21]

Which of the following diagrams best represents the equipotential lines around a negative point charge?



20. [NYJC/H2/Prelims 2018/P1/18]

An insulated rod with equal and opposite charges at its ends is placed in a non-uniform electric field with field lines as shown.

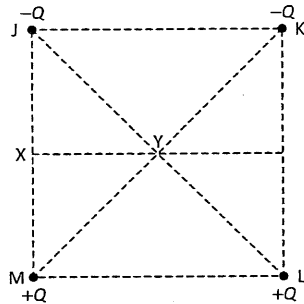


The rod experiences

- A a resultant force in the plane of the paper and a torque.
- B a resultant force in the plane of the paper but no torque.
- C a resultant force normal to the plane of the paper and a torque.
- D a resultant force normal to the plane of the paper but no torque.

21. [NYJC/H2/Prelims 2018/P1/19]

Four point charges are placed at the corners of a square JKLM. The point charges at J and K have a charge of $-Q$ and the point charges at L and M have a charge of $+Q$.



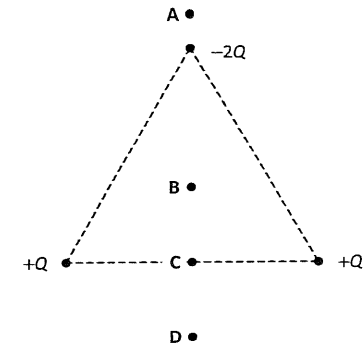
An electron is brought from point X to point Y along a straight path by an external force without any change in its speed.

Which of the following correctly describes the changes in the electric potential energy of the system and the work done by the external force?

	change in electric potential energy	work done by external force
A	positive	positive
B	negative	negative
C	negative	positive
D	zero	zero

22. [PJC/H2/Prelims 2018/P1/16]

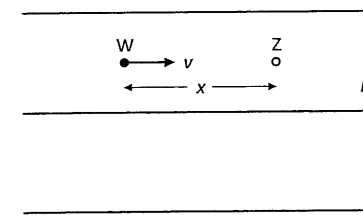
Three charges of charge $+Q$, $+Q$, and $-2Q$ rest at the corners of an equilateral triangle as shown. A small test charge $+q$ is brought near the three charges.



At which of the following positions would it be possible for the small test charge to experience zero net force?

23. [PJC/H2/Prelims 2018/P1/17]

An electron of charge $-e$ and mass m travels from point W to point Z within a uniform electric field of strength E . At point W, the particle has a velocity of v . It comes to a stop at point Z. The distance between point W and Z is x .

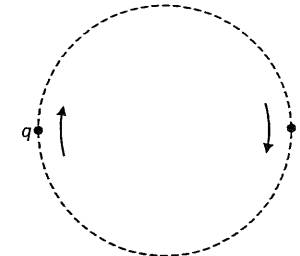


Which of the following gives the correct expression for x ?

- A $\frac{mv}{E}$
- B $\frac{mv}{eE}$
- C $\frac{mv^2}{2E}$
- D $\frac{mv^2}{2eE}$

24. [PJC/H2/Prelims 2018/P1/18]

Two charged particles, each of charge $q = 1.6 \times 10^{-19} \text{ C}$, are moving in a circular path in an applied magnetic field. The period of the motion for each particle is $2.0 \times 10^{-10} \text{ s}$.

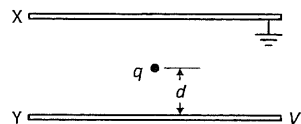


What is the current caused by the motion of the two particles?

- A $1.6 \times 10^{-29} \text{ A}$
- B $3.2 \times 10^{-29} \text{ A}$
- C $8.0 \times 10^{-10} \text{ A}$
- D $1.6 \times 10^{-9} \text{ A}$

25. [RVHS/H2/Prelims 2018/P1/19]

An oil drop of charge q and mass m is suspended at the midpoint of a pair of horizontal plates X and Y. Plate X is earthed, while plate Y is held at a potential of V with a total charge of Q_Y . The drop is at a distance d from plate Y.



What is the expression for the potential V ?

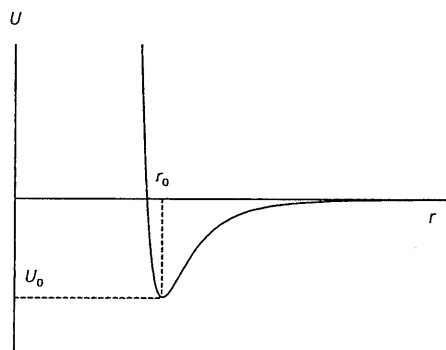
- A $\frac{mgd}{q}$ B $\frac{2mgd}{q}$ C $\frac{mg}{q}$ D $\frac{Q_Y}{4\pi\epsilon_0 d}$

26. [RVHS/H2/Prelims 2018/P1/20]

The simplified interaction between a pair of atoms can be represented by an equation relating the electric potential energy of the atoms,

$$U = \frac{A}{r^{12}} - \frac{B}{r^6}$$

where U is the electric potential energy of the atoms, r is the separation between them, and A and B are constants. The variation of U with r is shown below.



Which of the following statements is correct?

- A The atoms experience no interatomic forces at the separation of r_0 .
 B The atoms experience the strongest forces of attraction as r decreases from r_0 .
 C To separate the atoms, the electric potential energy of the atoms must decrease by a magnitude of U_0 .
 D As the separation between the atoms increases from r_0 , the magnitude of resultant force acting on the atoms increases then decreases.

27. [SRJC/H2/Prelims 2018/P1/21]

A negatively charged sphere P is balanced midway between two horizontal plates when a potential difference V is applied between the plates.

Which statement is correct?

- A Increasing V increases both the electric and the gravitational potential energy of the sphere.

- B Increasing V decreases the electric potential energy and increases the gravitational potential energy of the sphere.
 C Decreasing V decreases both the electric and the gravitational potential energy of the sphere.
 D Decreasing V increases both the electric and gravitational potential energy of the sphere.

28. [SRJC/H2/Prelims 2018/P1/22]

A current of 8.0 A passes through a cylindrical copper wire with a diameter of 8.0 mm. The density of copper is 8960 kg m^{-3} and the mass of a single copper atom is $1.0 \times 10^{-25} \text{ kg}$.

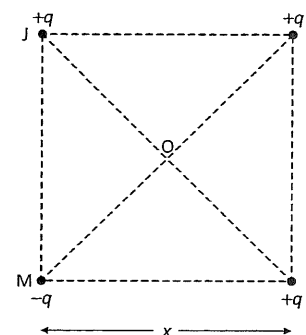
Assuming that there is one conduction electron for each copper atom, what is the drift velocity of the electrons in the wire?

- A $2.8 \times 10^{-6} \text{ m s}^{-1}$ B $3.4 \times 10^{-6} \text{ m s}^{-1}$ C $1.1 \times 10^{-5} \text{ m s}^{-1}$ D $5.8 \times 10^{-5} \text{ m s}^{-1}$

29. [TPJC/H2/Prelims 2018/P1/21]

Four point charges, each of magnitude q , lie at the four corners J, K, L, M of a square.

The signs of the charges are shown in the diagram. The side of the square is of length x .



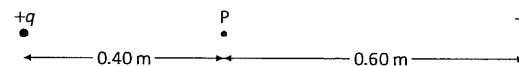
What is the electric field strength at the centre O of the square?

- A $\frac{q}{\pi\epsilon_0 x^2}$ towards K B $\frac{q}{\pi\epsilon_0 x^2}$ towards M
 C $\frac{q}{2\pi\epsilon_0 x^2}$ towards K D $\frac{q}{2\pi\epsilon_0 x^2}$ towards M

30. [VJC/H2/Prelims 2018/P1/18]

Two equal and opposite charges $+q$ and $-q$ are situated 1.00 m apart.

At a point P, 0.40 m from the charge $+q$, the magnitude of the electric field strength due to the charge $+q$ alone is 12 N C^{-1} .

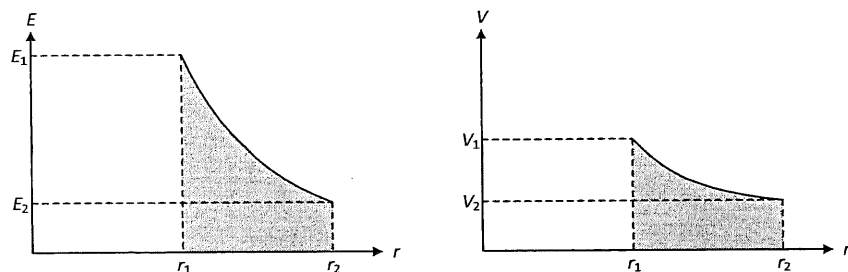


What is the magnitude of the electric field strength at P due to both charges?

- A 17 N C^{-1} B 20 N C^{-1} C 23 N C^{-1} D 30 N C^{-1}

31. [YJC/H2/Prelims 2018/P1/19]

The two graphs below represent the variation with distance r of the electric field strength and the electric potential around an isolated point charge. The electric field strengths (E_1 and E_2) and electric potentials (V_1 and V_2) at two distances r_1 and r_2 are indicated in the diagram.

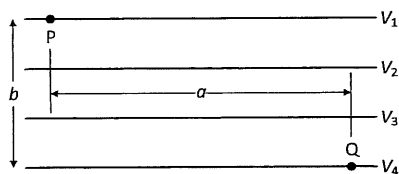


The work done by an external force in moving a test charge $+q$ from $r_2 = 2r$ to $r_1 = r$ is equal to

- A $q \times \text{shaded area under } E\text{-}r \text{ graph}$
- B $q \times \text{shaded area under } V\text{-}r \text{ graph}$
- C $q \times \frac{1}{2}(E_1 + E_2)$
- D $q \times (V_2 - V_1)$

32. [YJC/H2/Prelims 2018/P1/18 Modified]

An electron, placed at a point P inside a uniform electric field, experiences a force of magnitude F . The horizontal lines are equipotential lines with $V_4 > V_1$.



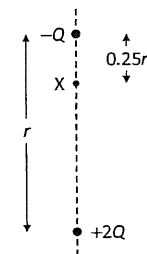
The electron is moved from P to Q.

What is the work done on the electron by the electric field?

- | | |
|---------|---------|
| A $-Fb$ | B $-Fa$ |
| C Fb | D Fa |

33. [YJC/H2/Prelims 2018/P1/19]

Two point charges of charge $-Q$ and $+2Q$ are placed along a line as shown. The distance between them is r .



What is the change in electric potential energy of the system when a point charge of charge $+3Q$ is placed at position X, a distance $0.25r$ from $-Q$?

- | | | | |
|----------------------------------|---------------------------------|-----------------------------------|----------------------------------|
| A $\frac{5Q^2}{\pi\epsilon_0 r}$ | B $\frac{Q^2}{\pi\epsilon_0 r}$ | C $-\frac{5Q^2}{\pi\epsilon_0 r}$ | D $-\frac{Q^2}{\pi\epsilon_0 r}$ |
|----------------------------------|---------------------------------|-----------------------------------|----------------------------------|

II. Structured Questions (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P2/4]

A uniformly charged non-conducting sphere of radius R has a positive charge of $+q$ is shown in Fig. 1.1. A second uniformly charged non-conducting sphere, which has a negative charge of $-q$, is brought close to the first sphere, such that the distance between their centres is d . The second sphere has a smaller radius r .

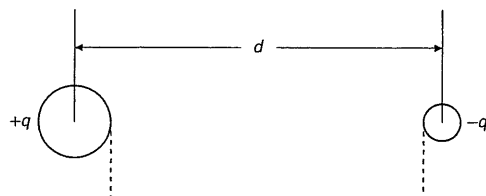


Fig. 1.1

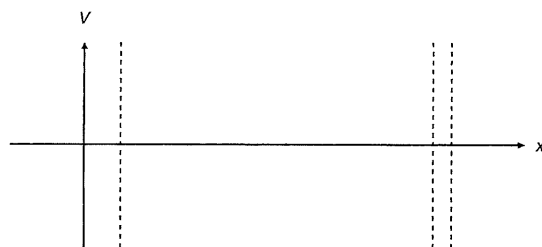


Fig. 1.2

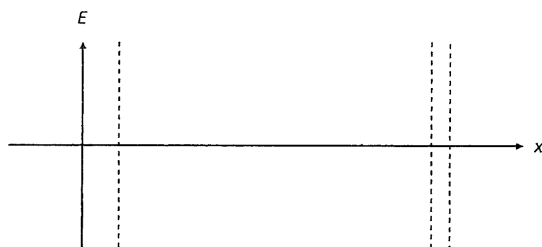


Fig. 1.3

- Sketch on Fig. 1.2 the variation with distance x from the centre of the positively charged sphere of the electric potential V between the surfaces of the charged spheres, due to
 - the charge on the positively charged sphere only. Label this graph X.
 - the charge on the negatively charged sphere only. Label this graph Y.
 - the charges on both the positively and negatively charged spheres. Label this graph Z.
- Hence, using Fig. 1.2, sketch on Fig. 1.3 the variation with x of the resultant electric field strength E between the surfaces of the charged spheres.

- The negatively charged sphere is replaced with a charge of $-2q$. A positive point test charge remains stationary when placed at a point along the axis joining the centres of the two spheres. Determine the distance between this point and the centre of the sphere with $+q$ charge in terms of d .

[Total: 7]

2. [AJC/H2/Prelims 2018/P3/7 Part]

- Define *electric field strength* at a point.
- A small charged metal sphere is situated at the centre of an earthed metal box. Fig. 2.1 illustrates the electric field between the sphere and the box.

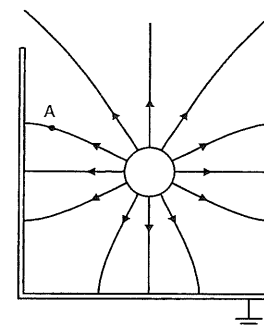


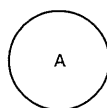
Fig. 2.1

- By reference to Fig. 2.1, state and explain
 - whether the sphere is positively or negatively charged,
 - why it appears as if the charge on the sphere is concentrated at the centre of the sphere.
- On Fig. 2.1, draw an arrow to show the direction of the force on a stationary electron situated at point A.
- The radius of the sphere is 2.4 cm. The magnitude of the charge on the sphere is 0.76 nC.
 - Calculate a value for the magnitude of the electric potential on the surface of the sphere.
 - State the sign of the charge induced on the inside of the metal box. Hence, explain whether the actual magnitude of the potential on the surface of the sphere will be greater or smaller than the value calculated in b.iii.1.
- Two electrons on the surface of the metal box are separated by a distance of 6.0×10^{-10} m. Determine the ratio of the electric force between the electrons to the weight of an electron.
- A lead sphere is placed in a lead box, in a similar arrangement to that shown in Fig. 2.1. Explain why it is not possible for the gravitational field to have a similar shape to that of the electric field.

[Total: 11]

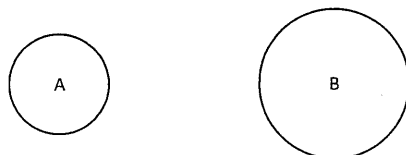
3. [CJC/H2/Prelims 2018/P3/8 Part]

- c. i. Explain what is meant by an *electric field*. [1]
- ii. Draw the charge distribution and the electric field lines around the following bodies:
1. An isolated positively charged metal sphere A.



[2]

2. When a larger, neutral metal sphere B is brought close to the positively charged metal sphere A in i.



[4]

[Total: 7]

4. [DHS/H2/Prelims 2018/P2/5 Modified]

- a. i. State what are meant by the terms *electric field* and *electric field strength*. [2]
- ii. Determine the electric field strength at a distance of 25 cm from a point charge of $5.2 \times 10^{-7} \text{ C}$. Give a unit for electric field strength with your answer. [3]
- b. Fig. 4.1 shows three charges of charge $1.0 \mu\text{C}$ at X, $-1.0 \mu\text{C}$ at Y and $1.0 \mu\text{C}$ at Z. These charges are at the corners of an equilateral triangle.

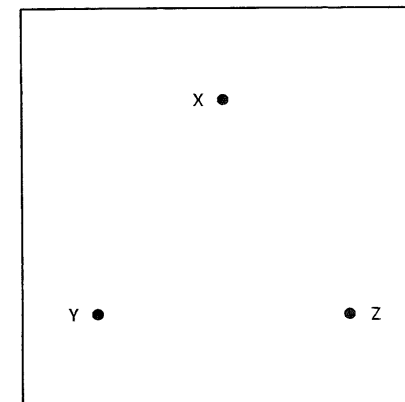


Fig. 4.1

Without making any calculation, draw on Fig. 4.1 the electric field within the given rectangular area, indicating its main characteristics. [4]

[Total: 8]

5. [EJC/H2/Prelims 2018/P3/3]

- a. Define *electric potential* at a point. [1]
- b. Two positive point charges of magnitude e are set up along the line XY as shown in Fig. 5.1.

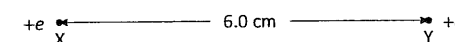


Fig. 5.1



Fig. 5.2

- i. On Fig 5.2, sketch the variation of the electric potential V between X and Y. [2]
- ii. Describe how the electric force acting on a known charge q placed at a point along XY could be obtained from Fig 5.2. [2]
- c. i. A negative point charge of charge $-e$ is now placed at the midpoint of XY. State the magnitude of electric force experienced by the point charge. [1]
- ii. The negative point charge at the midpoint in XY in c.i. is moved to the position Z as shown in Fig 5.3.

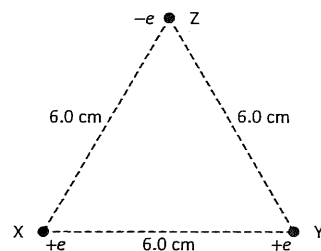


Fig. 5.3

Within the bounded region of Fig 5.4, sketch the resultant electric field lines of the 3 point charges. [3]

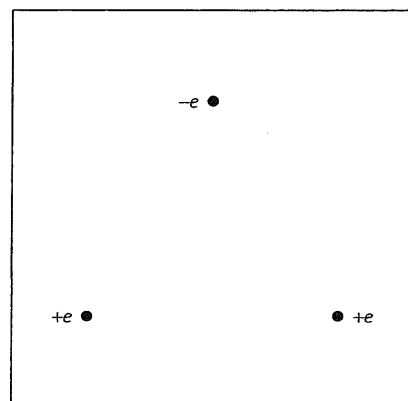


Fig. 5.4

- iii. Determine the work done by the electric field to completely separate the 3 point charges in Fig 5.3. [3]

[Total: 12]

6. [H1/H2/Prelims 2018/P2/4 Modified]

- a. State what is meant by an electric field. [1]
- b. Fig. 6.1 below shows a set of equipotential lines of a region of an electric field.

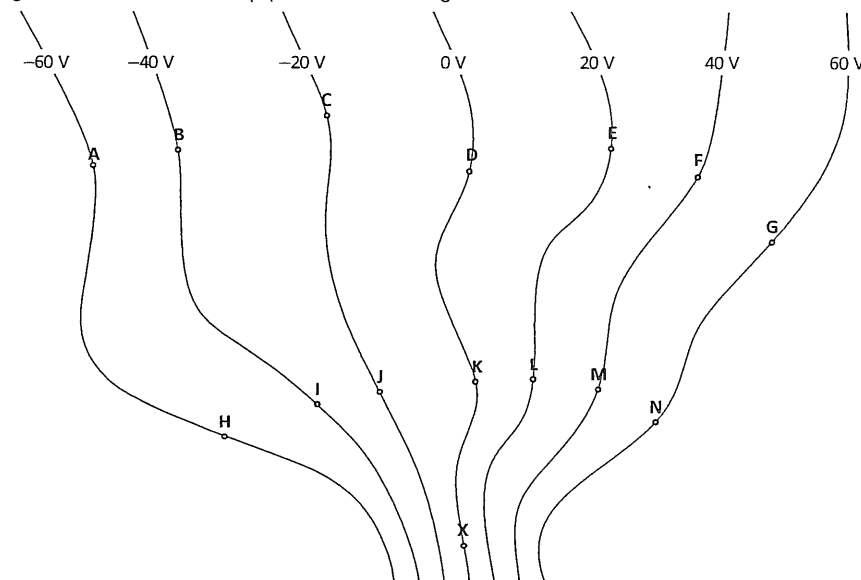


Fig. 6.1 (drawn to scale)

- i. Given that the same field line passes through the points A, B, C, D, E, F and G, draw in Fig. 6.1, this field line, clearly indicating its direction. [2]
- ii. Explain whether the field is stronger at D or at X. [1]
- iii. A charge Q_1 of $-5.0 \mu\text{C}$ is placed at the point L. Calculate the electric potential energy of the charge Q_1 at point L. [2]
- iv. Charge Q_1 is now released from rest. Given that work done on it by the electric field is $1.0 \times 10^{-4} \text{ J}$, identify the point that can represent the final location of Q_1 . [2]
- v. Charge Q_1 is now removed, and Q_2 of $-15.0 \mu\text{C}$ is now placed at the point X. By making direct measurements from the Fig 6.1, determine the magnitude of the electric force experienced by the charge. Indicate on Fig. 6.1 the direction of the force experienced by Q_2 at X. [3]

[Total: 11]

7. [JC/H2/Prelims 2018/P2/5 Modified]

A small sphere has a charge q of $+1.11 \times 10^{-6} \text{ C}$.

- Determine the number of electrons that have been removed from the sphere. [2]
- On the graph in Fig 7.1, sketch the variation of the electric field strength E between $x = 0.50 \text{ m}$ and $x = 2.00 \text{ m}$ from the centre of the sphere. The values at 0.50 m and 2.00 m are already shown. [2]

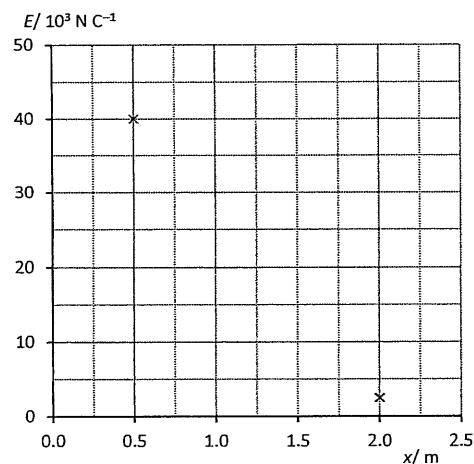


Fig. 7.1

- Another identical sphere, also of charge $+1.11 \times 10^{-6} \text{ C}$, is brought from a distant point to a distance of 1.2 m from the first sphere as shown in Fig. 7.2.

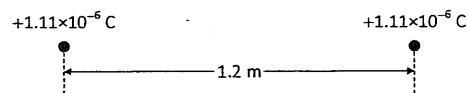


Fig. 7.2

- Determine the work done to bring the second sphere into position. [2]
- Determine the magnitude and direction of the electric field strength at the point between the two spheres that is 0.70 m from the sphere on the left of Fig. 7.2. [3]

[Total: 9]

8. [JC/H2/Prelims 2018/P3/5]

- Define *electric field strength*. [1]
- Electrons from a hot filament enter a region between a pair of horizontal, parallel plates after being accelerated by an electric field. Fig. 8.1 shows the electrons travelling horizontally at a speed of $2.50 \times 10^7 \text{ m s}^{-1}$ before entering the plates.

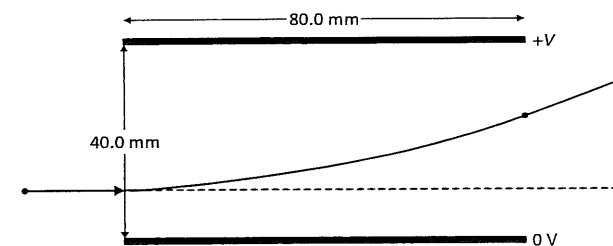


Fig. 8.1

- The electrons deviate by an angle of 30° on leaving the parallel plate of 80.0 mm long. The separation between the plates is 40.0 mm . Calculate the time taken for the electrons to travel through the plates. [1]
- Calculate the vertical component of the velocity when the electrons exit the parallel plates. [2]
- Hence, calculate the acceleration of the electrons. [1]
- Calculate the potential difference between the two plates. [2]
- If the plate separation is reduced, suggest how the answer to b.ii. is affected. [1]

[Total: 8]

9. [MI/H2/Prelims 2018/P2/5]

- a. Define *electric field potential* at a point in an electric field. [1]
- b. Two spherical charges P and Q, each of mass 0.20 g, are separated by a horizontal distance of 30 mm as shown in Fig. 9.1. P has a charge of $+20\ \mu\text{C}$ and Q has a charge of $-40\ \mu\text{C}$.

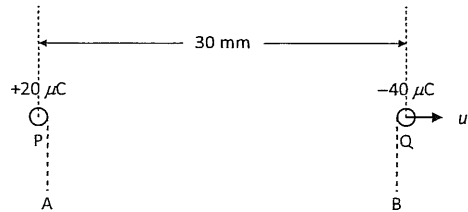


Fig 9.1

- i. In Fig. 9.2, r represents the distance from the centre of P to the centre of Q. On the axis provided, sketch the variation of the net electric potential V between boundaries A and B. [3]

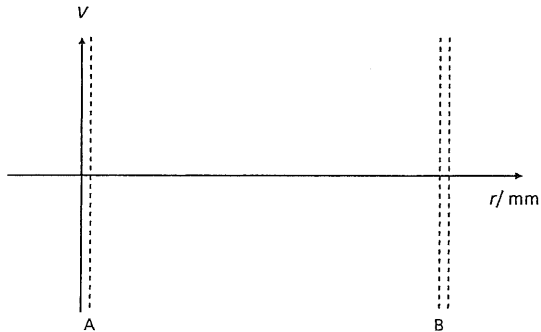


Fig. 9.2

- ii. Q is then projected away from P with an initial speed u such that it reaches a position far enough beyond the influence of charge P.
 - 1. Determine the electric potential energy of Q at its initial position. [2]
 - 2. Determine the minimum initial speed u required. [2][Total: 8]

10. [MJC/H2/Prelims 2018/P3/6 Modified]

Fig. 10.1 shows an isolated charged conducting sphere with equipotential lines (dashed lines).

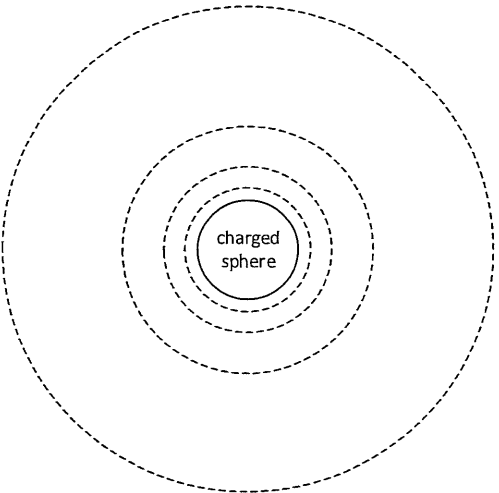


Fig. 10.1

For points on the surface or outside the sphere, the charge on the sphere behaves as if it were concentrated at the centre.
The values of the electric potential V at a distance x from the centre of the sphere are shown in Fig. 10.2. These values do not correspond to the equipotential lines in Fig. 10.1.

x/m	$V/10^5\text{ V}$
0.19	-1.50
0.25	-1.14
0.32	-0.89
0.39	-0.73

Fig. 10.2

- a. On Fig. 10.1, draw the electric field lines. Label these lines E . [2]
- b. Explain how your drawing in a. shows the relationship between electric potential V and the electric field strength E . [2]
- c.
 - i. Use the data in Fig. 10.2 to show that the potential V is inversely proportional to the distance x . Explain your reasoning. [2]
 - ii. The potential at the surface of the sphere is $-1.9 \times 10^5\text{ V}$. Calculate the radius of the sphere. [2]
 - iii. Determine the charge on the sphere. [2]

[Total: 10]

11. [NJC/H2/Prelims 2018/P2/6 Modified]

- a. A dust particle is suspended in the air between two parallel plates with the direction of electric field shown in Fig. 11.1. The electric field strength between the two parallel plate is $3.0 \times 10^3 \text{ V m}^{-1}$. The particle has charges of $+1.4 \times 10^{-15} \text{ C}$ and $-1.4 \times 10^{-15} \text{ C}$ near its ends. The charges may be considered to be static point charges separated by a distance of 3.0 mm. The particle makes an angle of 35° with the direction of the electric field.

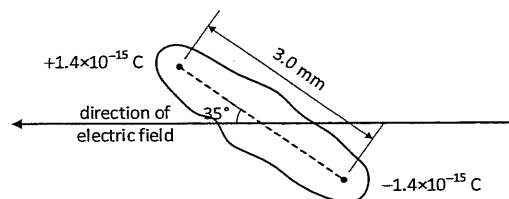


Fig. 11.1

- Calculate the magnitude of the force on each charge due to the electric field. [1]
 - Describe and explain the subsequent motion of the particle in the electric field. [3]
- b. Fig. 11.2 and Fig. 11.3 show the electric field pattern near two point charges respectively.

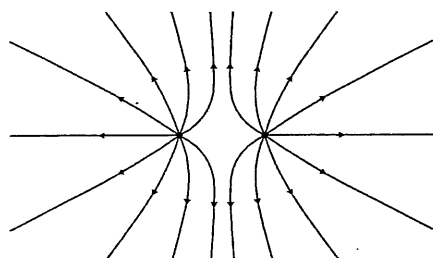


Fig. 11.2

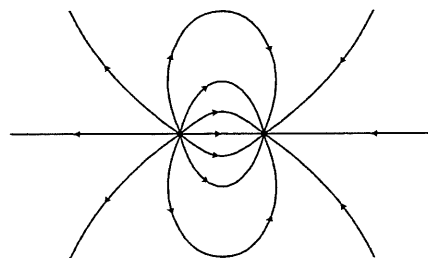


Fig. 11.3

- On the appropriate figure(s), mark the point(s)
 - with zero electric field and label it with the letter "N"
 - with zero electric potential and label it with the letter "E"
 - On Fig. 11.2, draw three complete equipotential lines. [2]
- c. Fig. 11.4 shows the variation with distance r of the electric potential V and the magnitude of the electric field strength E for an isolated -1.0 C charge.

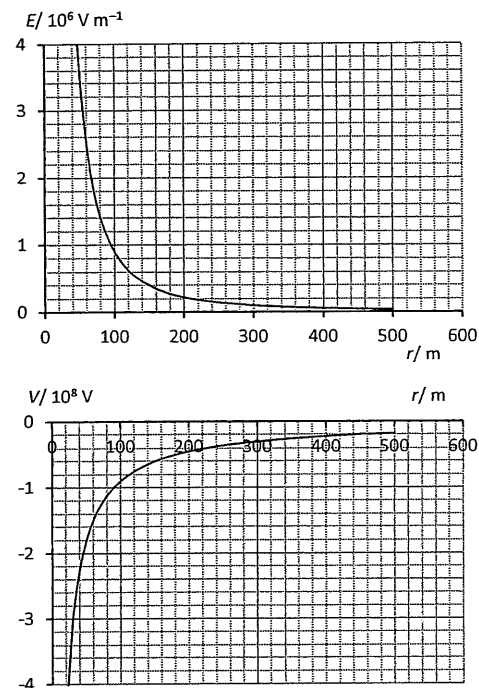


Fig. 11.4

Using the values from Fig. 11.4, state and explain two relationships between the graphs. [4]
[You will may annotate on the figure and make clear reference to these annotations to support your answer.]

[Total: 11]

12. [PJC/H2/Prelims 2018/P3/8 Modified]

- Define *electric field strength* at a point. [1]
- Fig. 12.1 shows two horizontal plates A and B whose electric potentials are 80.0 V and -20.0 V , respectively. The plates are 90.0 cm apart. A conductor P is situated centrally between the two plates.

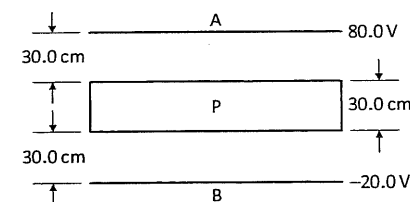


Fig. 12.1

- i. On Fig. 12.2, sketch the variation with distance x from A of the electric potential V .

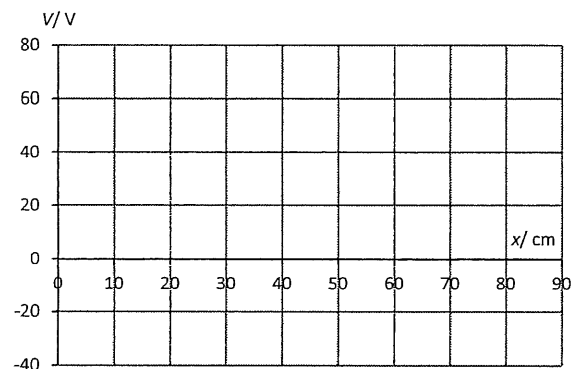


Fig. 12.2

- ii. On Fig. 12.3, sketch the variation with distance x from A of the electric field strength E .

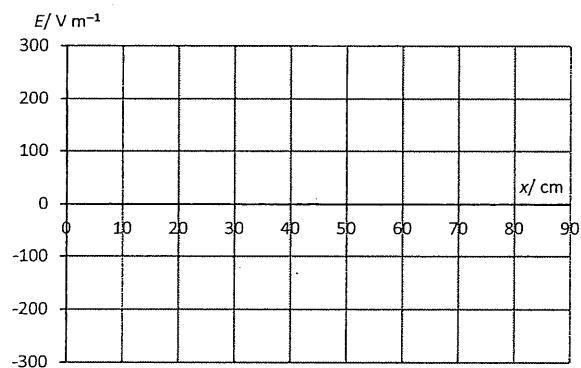


Fig. 12.3

- c. Fig. 12.4 shows an electron at rest in the centre between two vertical plates A and B. The length of the plates is 20.0 cm and they are 90.0 cm apart.

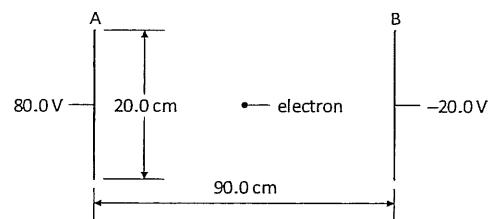


Fig. 12.4

- i. Describe the path travelled by the electron between the two plates.
- ii. Determine the horizontal speed at which the electron would hit one of the plates.

[1]

[2]

[Total: 8]

13. [RVHS/H2/Prelims 2018/P2/6]

In a cathode ray oscilloscope, an electron beam is fired from rest with an accelerating voltage of 1.0 kV and strikes a fluorescent screen that is 14.0 cm from the end of the vertical deflection plates as shown in Fig. 13.1. Depending on the potential difference between the horizontal and vertical deflection plates, the beam can be manipulated to draw waveforms on the screen.

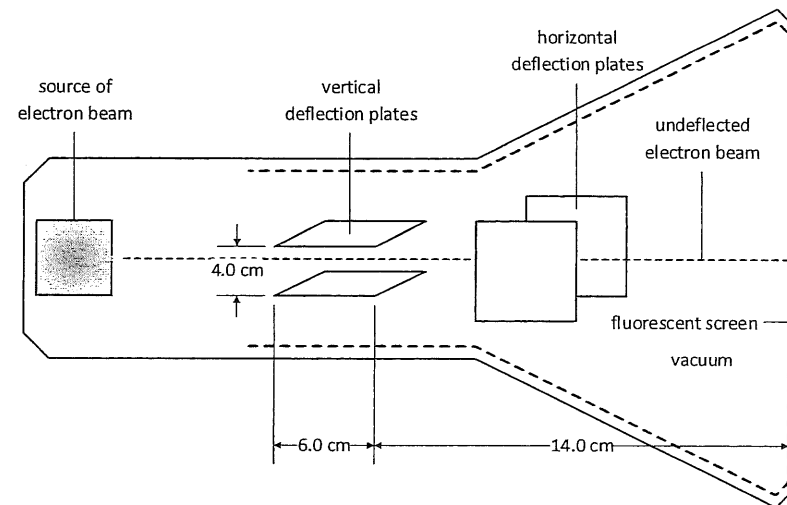


Fig. 13.1 (not to scale)

An alternating voltage is applied across the vertical deflection plates. The time-base is set at 5.0 ms per division and the vertical scale of the display is set at 5.0 V per division. The waveform on the screen is shown in Fig. 13.2.

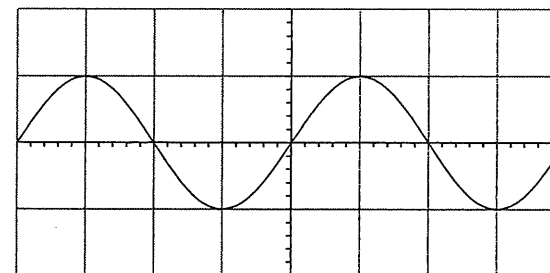


Fig. 13.2 (not to scale)

- a. Calculate the speed of the electrons entering the region between the vertical deflection plates.
- b. Show that the time taken for an electron to travel between the vertical deflection plates is 3.2 ns.
- c. Determine the maximum vertical deflection of the electron beam when it reaches the fluorescent screen if the potential difference between the vertical deflection plates is 20 times that of the maximum reading in Fig. 13.2.
- d. Describe and explain how the waveform in Fig. 13.2 would be different if the electron beam was replaced with a proton beam.

[2]

[1]

[4]

[2]

[Total: 9]

14. [SRJC/H2/Prelims 2018/P3/4 Modified]

- a. Define *electric potential* at a point. [2]
- b. Two charged solid spheres A and B are situated in a vacuum. Their centres are separated by a distance of 30.0 cm, as shown in Fig. 14.1.

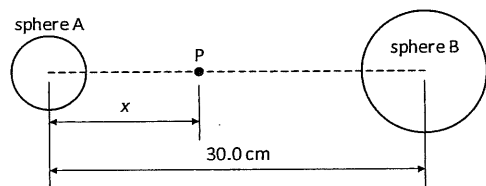


Fig. 14.1 (not drawn to scale)

The variation with distance x from the centre of sphere A of the electric potential V_A and V_B due to sphere A and B independently is shown in Fig. 14.2.

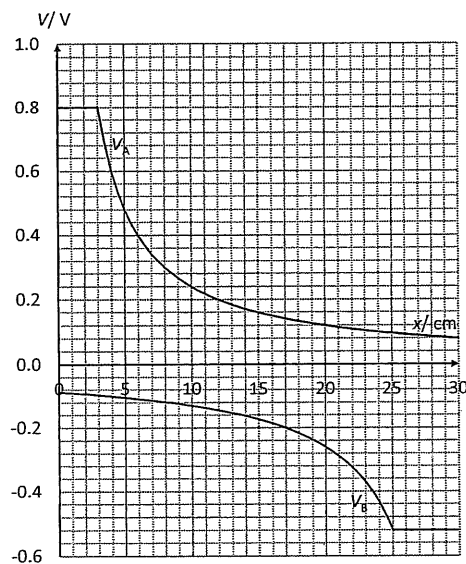


Fig. 14.2

- i. Using Fig. 14.2, state the radius of both spheres. [1]
- ii. State and explain the signs of both spheres. [2]
- iii. Point P is at a distance $x = 10.0$ cm. An alpha particle has kinetic energy E_k when at infinity. Use Fig. 4.2 to determine the minimum value of E_k such that the alpha particle may travel from infinity to point P. [3]

[Total: 8]

15. [TPJC/H2/Prelims 2018/P3/4]

- a. State what is meant by an *electric field*. [1]
- b. Fig. 15.1 shows a small charged particle placed at point A in a uniform electric field. The particle experiences an electric force F of 5.0×10^{-7} N. The grid lines are at intervals of 1.0 cm.

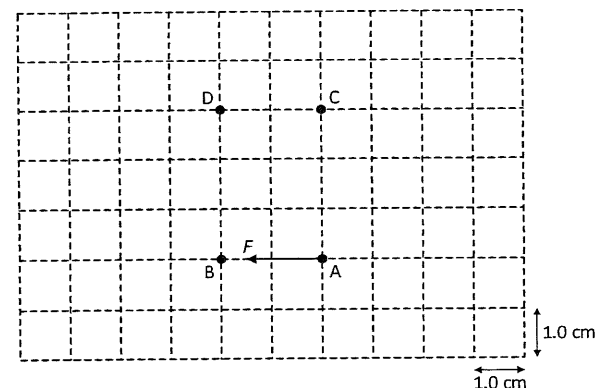


Fig. 15.1

Determine the work done by the electric force F if the particle moves

- i. from A to B, [1]
- ii. from A to D. [1]
- c. If the particle carries a charge of -2.5×10^{-11} C and the potential at point A is +200 V, calculate the potential at point B in the uniform field. [2]
- d. The uniform electric field is produced by a pair of flat, parallel metal plates, one of which is earthed and the other is at a potential of +1000 V. On Fig. 15.1, sketch
- i. the positions of the two plates. [2]
- ii. at least four equipotential lines in the region between the two plates. Label the value of the potential of each line that is drawn. [2]

[Total: 9]

16. [VJC/H2/Prelims 2018/P3/5]

A beam of electrons travelling at a speed of $6.8 \times 10^6 \text{ m s}^{-1}$ and making an angle of 30° to the horizontal enters the region between two horizontal parallel plates at point P. The electrons are deflected and exit horizontally at point Q as shown in Fig. 16.1. The length of the plates is L and their separation is 10 cm. The potential of the lower plate is +45 V with respect to the upper plate.

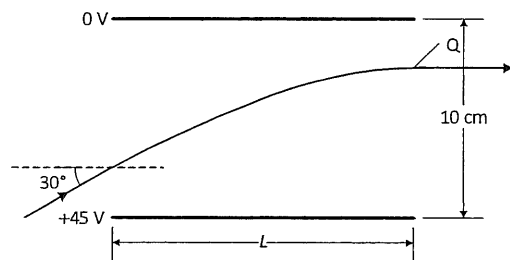


Fig. 16.1

- Show that the magnitude of the acceleration of an electron between the plates is $7.9 \times 10^{13} \text{ m s}^{-2}$. [2]
- Determine the length L of the plates. [3]
- Charged particles travelling horizontally at a speed of $6.8 \times 10^6 \text{ m s}^{-1}$ enter the region between the plates at point P as shown in Fig. 16.2.

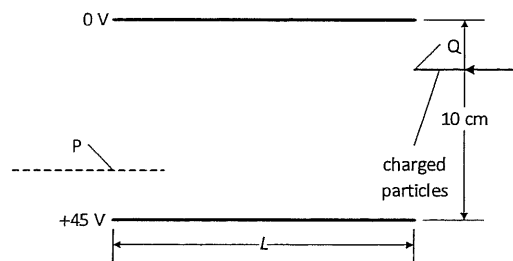


Fig. 16.2

- On Fig 16.2, sketch the path of the charged particles between the plates if they were electrons. Label this path A. [1]
- On Fig 16.2, sketch the path of the charged particles between the plates if they were protons. Label this path B. [1]
- Explain two differences between paths A and B. [2]

[Total: 9]

17. [YJC/H2/Prelims 2018/P3/5 Modified]

- Two point charges of $-6.0 \mu\text{C}$ and $+6.0 \mu\text{C}$ are arranged at points A and B respectively as shown in Fig. 17.1.

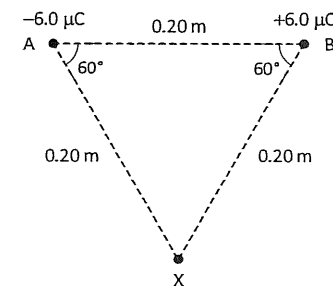


Fig. 17.1

Indicate clearly on Fig. 17.1 the directions of

- the electric field at X due to the charge at A (label it E_A), [1]
 - the electric field at X due to the charge at B (label it E_B), and [1]
 - the resultant electric field at X due to both charges (label it E_R). [1]
- Determine the magnitude of the resultant electric field strength at X. [2]

[Total: 5]